

inline: Streamlining restaurant booking management for better dining experiences

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ABOUT inline



About inline

To address restaurant needs for efficient management and customer service, inline uses the highly available tools of Google Cloud to swiftly develop user-reliable features.

Founded in 2015, inline is a Taiwanese software provider that offers digital solutions to help restaurants manage their tables, reservations and in-person queues efficiently. Serving more than 7000 restaurants and 200 million users across Taiwan, Hong Kong, Singapore, Australia and Japan, inline constantly adds new features to its restaurant management system to meet its customers' needs.

Industries: Food & Beverage

reducing mainter
workloads.

Location: Taiwan

Google Cloud results

- Helps develop and operate a new mobile app with a small team of five by removing infrastructure management workloads
- Supports cross-device data synchronization for better restaurant table management with Firebase
- Enables quick development of food ordering and delivery features to meet restaurants' business needs

Enables
search
online
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flow

(<https://cloud.google.com/kubernetes-engine/>),
Google Cloud (<https://cloud.google.com/>) query/),
Google Maps Platform

Cloud Functions
(<https://cloud.google.com/functions/>)

Cloud SQL (<https://cloud.google.com/sql/>)

Firebase (<https://firebase.google.com/>) to help.

Kubernetes Engine
(<https://cloud.google.com/kubernetes-engine/>)

BigQuery (<https://cloud.google.com/bigquery/>)

Google Maps Platform
(<https://mapsplatform.google.com/>)

Restaurants are always striving to provide better dining experiences for their customers. To achieve that, they need to offer not only delectable dishes and an exquisite ambiance, but also impeccable services including efficient table and booking management. After all, if consumers have to stand in line for hours for a table or find out their reserved tables are taken by others due to management negligence, even the finest cuisine could lose its charm.

[inline](https://inline.app/) (<https://inline.app/>) understands restaurants' need to manage their tables, reservations and in-person queues efficiently. Dedicated to delivering the best digital solutions, the Taiwanese tech startup released a queueing management system in 2016. Since its inception, it has quickly expanded it into a comprehensive restaurant management system covering reservations, tables, food ordering and food delivery. The software is currently available in Taiwan, Hong Kong, Singapore, Australia and Japan, serving more than 7000 restaurants and 200 million users.

"We always stand with our customers by addressing the issues they encounter when providing restaurant services. This is why we want our team to stay agile and flexible with the help of technology," says Iris Lu, digital marketing manager at inline. "For example, when restaurants in Taiwan were restricted from offering indoor dining due to COVID-19 in 2021, we

quickly launched the food delivery feature that enabled our customers to continue their businesses."

As a startup, inline had limited resources when it first started developing its restaurant management system. The company needed a powerful app development tool that can simplify development workloads while delivering excellent performance.

To that end, inline chose Firestore

(<https://firebase.google.com/firebase-and-gcp>) on Google Cloud (<https://cloud.google.com/>) to develop and deploy its software, because the team can easily test and run its app on Firebase without requiring a large team or resources to manage its infrastructure.

"When we started developing our restaurant management system, the idea was quite innovative, and we knew we would go through lots of trial-and-error along the development process," notes Spirit Tu, lead developer at inline. "Firestore on Google Cloud is the best choice for us because it enables us to develop an app in the most effortless way."

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**Supporting quick app
development and streamlined
restaurant management with
Firebase**

In the first two years after inline launched its mobile app for restaurant management, the company relied on a small team of five engineers to develop new features and oversee system operations. Since Firebase enables the team to develop and scale its app without the need to set up and manage servers, inline managed to build and run a well-functioning app that covers all aspects of table and booking management with limited personnel.

On top of that, Firebase supports cross-device data synchronization, which is very helpful for inline, because it could quickly realize the feature of displaying the latest booking status on a diner's multiple mobile devices. When a restaurant employee is reserving a table for a customer on one tablet, an alert immediately pops up on other tablets indicating that the table is being taken, which prevents double reservations.

"The cross-device data synchronization feature of Firebase is very important to our service, because restaurants can effortlessly avoid making double bookings and provide better dining experiences with it," says Casper Chen, program manager at inline. "Without Firebase, it would have taken us at least three more months to develop the same feature."

Although inline's app supported by Firebase is serverless, the company can still offer features that require heavy computing by connecting Firebase with Cloud Functions (<https://cloud.google.com/functions>). For example, on inline's mobile app, restaurant employees can see the most updated queue sequence calculated based

on guest numbers. The real-time calculation is realized by Cloud Functions, which retrieves queueing event data like cancellations and new reservations from Firebase, runs algorithms to generate the latest queue sequence numbers for each waiting guest group, and sends the numbers back to Firebase. This way, inline's customers are able to manage guest queues efficiently and seamlessly.

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Providing seamless online restaurant bookings with GKE

inline's restaurant management system also supports online reservation, and the company uses Kubernetes clusters on [Google Kubernetes Engine](https://cloud.google.com/kubernetes-engine) (<https://cloud.google.com/kubernetes-engine>) (GKE) as the servers of this service. Many of inline's customers are famous restaurants that open online reservations at certain times like the beginning of a month, which often leads to instant flow surges as high as 50X. Thanks to the autoscaling feature of GKE, which can automatically scale computing resources up and down according to actual needs, inline is able to ensure smooth online booking experience while reducing infrastructure costs.

In some extreme cases, the inline team found that the autoscaling feature of GKE couldn't scale up clusters as fast as needed. As Kubernetes is designed for customized use, inline's engineers were able to insert a tool that supports automatic pre-scaleup for all foreseeable traffic surges. As a result,

inline's online reservation service on GKE has never encountered any disruption due to high traffic.

Developing high-performance and well-integrated app features with a single cloud ecosystem

Before COVID-19 escalated to a worldwide pandemic, the inline team knew that it had to prioritize developing the features that support takeaway and food delivery services as fast as possible, so its customers could be well prepared for potential restrictions on indoor dining.

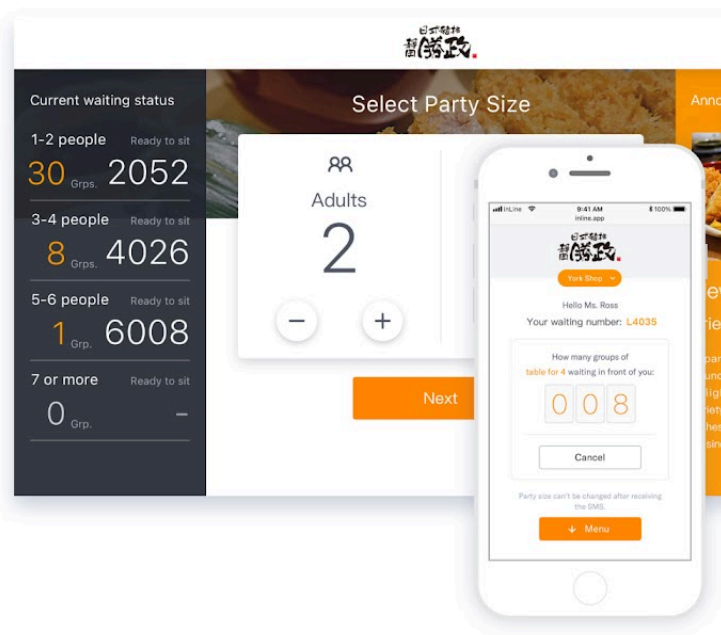
By leveraging Google Maps Platform

(<https://mapsplatform.google.com/>) to calculate delivery distance, the inline team was able to complete the development of the new features in less than three months and have them well-integrated into its existing system on Google Cloud. inline's restaurant customers in Taiwan were therefore able to continue their businesses without a hitch when indoor services were restricted later in 2021.

For inline, adding more Google Cloud products to support its services means not only shorter development time and better integration, but also higher system performance. When building the takeaway and delivery features, inline wanted to deploy the database outside Firebase to lower risks of total system failure by diversifying the data source of its services. The company eventually chose to use Cloud SQL (<https://cloud.google.com/sql>), because compared to the database products of

other public cloud platforms, Cloud SQL delivers shorter delays and prevents data loss by transferring data through the intranet of Google Cloud.

"We prefer to use products in the same cloud ecosystem for DevOps convenience and better system performance. With the abundant and powerful tools provided by Google Cloud, we're able to stay on one cloud platform as much as possible and deliver excellent services in a timely manner," says Tu.



inline app

Leveraging data analytics to serve up better service quality

Moving forward, inline hopes to help restaurants enhance their service quality by analyzing customers' business data. Lu says that thanks to

BigQuery (<https://cloud.google.com/bigquery/>), the company was able to produce data reports more efficiently for faster decision making. In the future, inline plans to help restaurants to understand diners' preferences better to provide personalized services through more in-depth and comprehensive analyses.

"With Google Cloud, we have managed to help our customers greatly improve their services and management efficiency by quickly developing features that address their needs. We will continue optimizing and expanding our services while exploring more potential applications of Google Cloud," she adds.

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—*Iris Lu, Digital Marketing
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